# CO3011

# SOFTWARE PROJECT MANAGEMENT

## PROJECT ASSIGNMENT

*Managing a Small Software Project from Initiation to Delivery*

Lecturer: Nguyen Minh Tam

Academic year: 2026-2027

Team size: 4-5 students

Project duration: milestone dates announced on LMS

## Project Overview

This assignment requires each team to plan, develop, monitor, control, and deliver a small software product while applying software project management concepts throughout the project lifecycle. The software product provides a real setting in which students must make and justify project-management decisions. The emphasis is therefore not on building the largest or most technically sophisticated system, but on managing a realistic software project in a disciplined, traceable, and evidence-based manner.

Each team will receive a short project brief from the lecturer. Teams are expected to clarify the project, establish a baseline plan, develop the agreed product, and evaluate the final outcome.

## Learning Objectives

- Evaluate a proposed software project and establish clear objectives and measures of effectiveness
- Identify stakeholders, project authority, communication needs, standards, constraints, and project organization
- Apply the Step Wise approach to identify products, activities, estimates, risks, resources, checkpoints, and lower-level plans
- Distinguish effort from elapsed time and develop realistic cost and schedule estimates
- Identify and assess project and activity risks, and plan risk-reduction and contingency measures
- Allocate resources and revise plans to account for resource constraints
- Monitor actual progress against the project baseline and respond to deviations and change requests
- Deliver a working software product whose agreed scope can be traced to project-management decisions
- Review project performance and extract lessons learned from planned versus actual results

## Project Structure

The assignment is organized into three milestones and a final demonstration. Tasks are completed progressively; later work must remain consistent with earlier decisions. A team may revise its plan when justified, but changes must be documented rather than silently rewriting the original baseline.

| Milestone | Main Focus | Main Deliverables |
|---|---|---|
| 1 | Initiation & Definition | Client consultation, evaluation, objectives, stakeholders, infrastructure, lifecycle |
| 2 | Planning Baseline | Products/activities, estimates, risks, resources, schedule, quality plan |
| 3 | Execution, Monitoring & Control | Working product, actual data, change response, revised plan |
| Final | Delivery & Review | Final product, final report, lessons learned, demonstration/viva |

## Client Consultation and Requirement Discovery

Teams should not assume that all necessary requirements, priorities, constraints, or acceptance conditions will be provided automatically.

Assessment of client consultation considers:

- Relevance of questions to project scope, objectives, effort, risk, schedule, quality, or acceptance
- Prioritization of important uncertainties rather than large numbers of trivial questions
- Appropriate follow-up based on previous client responses
- Documentation of the information obtained
- Evidence that client information materially influenced requirements, scope, estimates, risks, or other project decisions.

### Client Interaction Log

Teams must maintain a log using at least the following information:

| Date | Team Question | Client Response | Impact on Project |
|---|---|---|---|
| Week 1 | How are users enrolled? | Example response | Requirement/scope decision |

## Task 1: Project Initiation, Evaluation & Definition

Students need to perform the following tasks:

Task 1.1. Analyze the initial project brief. Conduct client consultations and maintain the Client Interaction Log.

Task 1.2. Describe the organizational/business context, current problem or opportunity, proposed software product, major assumptions, and project constraints.

Task 1.3. Identify expected benefits and possible disbenefits, major costs, strategic/organizational contribution, and provide a justified go/no-go recommendation. Apply appropriate financial evaluation techniques where meaningful.

Task 1.4. Define the project scope, exclusions, major project objectives, expected outcomes, and measurable measures of effectiveness. Objectives should express intended project/organizational outcomes rather than merely restating deliverables.

Task 1.5. Identify the project authority, sponsor/client, project manager, decision responsibilities, and escalation mechanism.

Task 1.6. Identify major stakeholders and analyze their interests, influence, impact, and appropriate engagement strategy. Explain any modification to objectives, scope, or approach resulting from stakeholder analysis.

Task 1.7. Establish the project infrastructure: relevant standards/procedures, project team organization, roles/responsibilities, and communication methods.

Task 1.8. Analyze project characteristics, determine whether the project is primarily objective-based or product-based, identify high-level project risks, and select a suitable general development lifecycle/approach with justification.

**Expected outputs:** Client Interaction Log; project background and business case; evaluation and go/no-go recommendation; scope and exclusions; objectives and measures of effectiveness; project authority; stakeholder analysis; communication plan; standards and procedures; project-characteristics analysis; high-level risks; and lifecycle decision.

## Task 2: Product-Based Planning & Baseline

Task 2.1. Identify the major project products and construct a Product Breakdown Structure (PBS). Distinguish deliverable products from important intermediate products.

Task 2.2. Prepare Product Descriptions (PDs) for at least five important products. Each PD should address product identity, description, derivation, composition, format, relevant standards, and quality criteria.

Task 2.3. Construct a Product Flow Diagram (PFD) showing dependencies among the identified products. Identify specific product instances where appropriate.

Task 2.4. Derive the activities needed to create the products and construct an ideal activity network based on logical dependencies. Activity names should normally use a clear verb + noun form.

Task 2.5. Add appropriate checkpoints. For each important checkpoint, explain what is checked, who is responsible for review/approval, and what happens if the checkpoint is not passed.

Task 2.6. Carry out bottom-up effort estimates for major activities. Clearly distinguish effort from elapsed time and state important assumptions and the estimation method(s) used.

Task 2.7. Revise the activity structure to make activities controllable: decompose activities that are too long and bundle very short activities where appropriate.

Task 2.8. Identify and quantify meaningful activity-level risks. For important risks, estimate likelihood and impact, identify a risk owner, and specify both risk-reduction and contingency measures.

Task 2.9. Revise the project plan and estimates to include risk-response activities such as prototyping/pilot work, information gathering, or other appropriate measures.

Task 2.10. Identify and allocate resources to activities. Consider realistic availability, part-time participation, specialist constraints, non-project activities, and shared resources. Revise the ideal plan to produce a feasible resource-constrained schedule.

Task 2.11. Prepare the project schedule and cost baseline. Include activity dependencies, milestones, planned start/finish dates, Gantt chart, and critical path/network analysis where applicable.

Task 2.12. Review the quality aspects of the project plan, document the baseline, obtain appropriate team/client agreement, and establish Project Baseline V1.0.

**Expected outputs:** PBS; Product Descriptions; PFD; activity network; checkpoints; bottom-up estimates; cost estimate; activity-level risk register and responses; resource plan; resource-constrained schedule; Gantt chart; critical path/network analysis where applicable; quality arrangements; and Project Baseline V1.0.

## Task 3: Software Development, Monitoring & Control

Teams must implement the agreed software product during the execution period. The implementation scope is intentionally modest. The purpose is to generate authentic project-management evidence: actual effort, schedule deviations, quality issues, risks, resource problems, and change decisions.

Task 3.1. Implement the agreed core software scope. The final system should demonstrate coherent end-to-end workflows rather than a large collection of disconnected screens.

Task 3.2. Maintain a weekly project log recording planned work, actual work, planned effort, actual effort, issues/risks, and management actions.

Task 3.3. Monitor actual progress against Project Baseline V1.0. Record actual start/finish dates and actual effort for major activities. Explain significant deviations.

Task 3.4. Maintain and update the risk register. Record risks that materialize, newly identified risks, responses taken, and effects on schedule/cost/scope.

Task 3.5. Apply appropriate quality-control activities and maintain evidence that important agreed requirements/products have been verified.

Task 3.6. Respond to the project change/event issued by the lecturer. Analyze its effect on requirements/products, activities, effort, resources, risks, schedule, cost, and quality. Consider alternatives and justify the selected management response.

Task 3.7. Where the approved response changes the baseline, prepare a revised Project Baseline V2.0. Original baseline information must remain available for comparison.

Task 3.8. Create lower-level plans when additional detail becomes known during execution. Explain why the new detail was not appropriate or possible at the earlier planning stage.

**Expected outputs:** Working software increment/product; weekly project logs; planned-versus-actual data; updated risk/issue records; quality evidence; change-impact analysis; management decision; and revised Project Baseline V2.0 where applicable.

## Task 4: Final Delivery, Review & Closure

Task 4.1. Deliver and demonstrate the final agreed software product. Clearly identify completed, partially completed, deferred, and removed scope.

Task 4.2. Compare planned and actual project performance. Analyze effort, elapsed time, schedule, cost, scope, quality, and important risks. Explain major estimation errors and management decisions.

Task 4.3. Evaluate achievement of the original project objectives and measures of effectiveness. Distinguish product completion from achievement of project/business objectives.

Task 4.4. Prepare a project closure plan covering acceptance/handover, outstanding issues, release of resources, documentation, and post-project review where appropriate.

Task 4.5. Document lessons learned. Identify what the team would repeat, what it would change, and how the planning/management approach could be improved in a future project.

Task 4.6. Each group submits a concise individual contribution and reflection statement and must be prepared to answer individual viva questions.

**Expected outputs:** Final software product; final project-management report; baseline comparison; project-performance evaluation; closure/handover plan; lessons learned; individual reflection/contribution statement; and final demonstration.

## Mandatory Traceability

Project-management artifacts must form one coherent plan. Teams must maintain traceability among important objectives, products, activities, implementation, tests, risks, resources, and checkpoints. Creating independent tables that contradict one another will be penalized even when each table appears individually plausible.

| Objective | Requirement/Product | Activity | Resource | Risk | Verification/Test | Checkpoint |
|---|---|---|---|---|---|---|
| O1 | P1 | A1, A2 | Dev A | R3 | T1 | CP1 |

## Assessment and Grading

The project is graded primarily on the correctness, justification, consistency, realism, and demonstrated application of software project management concepts. A technically sophisticated product does not compensate for weak project management, and a polished management report does not compensate for failure to execute and deliver the agreed project.

| Assessment Component | Weight | Main Evidence |
|---|---|---|
| Client consultation, requirements & project definition | 10% | Interaction log; scope; objectives; measures |
| Project evaluation, stakeholders & infrastructure | 10% | Business case; authority; stakeholders; lifecycle |
| Product/activity planning | 12% | PBS; PD; PFD; activity network; checkpoints |
| Estimation, cost & scheduling | 12% | Effort/duration; cost; Gantt; dependencies |
| Risk management | 10% | Risk assessment; reduction; contingency; plan changes |
| Resource allocation | 8% | Allocation; constraints; feasible schedule |
| Software implementation & quality | 20% | Working agreed scope; quality/verification evidence |
| Monitoring, change management & replanning | 10% | Logs; actuals; change impact; revised baseline |
| Final review, closure & lessons learned | 4% | Planned-vs-actual; closure; reflection |
| Final demonstration & individual viva | 4% | Understanding, decisions, contribution |

### General Evaluation Criteria

| Criterion | What is evaluated |
|---|---|
| **Correctness** | Project-management concepts are applied correctly. |
| **Justification** | Important choices are supported by project evidence and clear reasoning. |
| **Consistency & Traceability** | Objectives, scope, products, activities, estimates, risks, resources, schedule, implementation, and tests agree with one another. |
| **Realism** | Assumptions, estimates, resources, risks, and responses are credible for the project context. |
| **Execution Evidence** | The team demonstrates that planning and management occurred during the project rather than being reconstructed only at the end. |

### Individual Accountability

The lecturer may adjust individual marks based on the individual viva, contribution evidence, project logs/repository history, and the student's ability to explain the team's artifacts and decisions. Every member is expected to understand the overall project, not only the part that they personally implemented.

## Submission Requirements

- Final project-management report in PDF format.
- Working link to the source-code repository and clear instructions for running/demonstrating the software.
- Project Baseline V1.0 and any approved revised baseline(s) must be retained for comparison.
- Client Interaction Log and weekly project logs must be included or linked as instructed.
- All major diagrams, estimates, schedules, risk registers, and traceability evidence must be readable.
- Intermediate milestone deadlines and exact submission instructions will be announced on the LMS.
- Final deadline: TBD.

## Final Demonstration and Viva

The final demonstration should be concise and prepared in advance. Teams should demonstrate the agreed end-to-end software workflows and, more importantly, the management evidence behind the project. The lecturer may compare the original baseline with actual project results and ask any team member to explain estimates, risks, resource decisions, schedule changes, client interactions, or scope decisions.

## Software Scope and Technical Expectations

The software product should be deliberately modest enough to complete during the assignment period. The project is not intended to reward unnecessary technical complexity. The minimum implementation should demonstrate the core agreed workflows, persistent data where appropriate, basic validation/error handling, and sufficient testing or verification evidence to support acceptance.

Additional features or advanced technologies do not automatically receive higher marks. Teams should prioritize the agreed project objectives and scope, manage available time and resources, and deliver a coherent product.

## Use of Generative AI

Generative AI may be used as a supporting tool for brainstorming, requirement refinement, estimation discussion, risk identification, coding assistance, debugging, testing, documentation, and similar activities. However, teams remain fully responsible for the correctness, consistency, quality, and understanding of all submitted work.

- The final report must declare the AI tools used and the tasks for which they were used.
- AI-generated questions, plans, estimates, risks, or code must be critically evaluated and adapted to the actual project.
- Students must be able to explain and defend all submitted artifacts and implementation during the viva.
- Fabricated client interactions, fabricated project data, or misleading declarations may be treated as academic-integrity issues according to course/university regulations.

## Important Notes

- Do not begin implementation based only on the initial one-line project brief. Clarify important uncertainties with the client.
- Do not silently modify old estimates or schedules after actual results are known. Preserve the original baseline and document approved revisions.
- More documentation is not automatically better. Concise, consistent, justified artifacts are preferred over large amounts of generic text.

- A project that encounters problems can still receive a strong mark if the team identifies, analyzes, communicates, and manages those problems appropriately.
- A working product alone is not sufficient for a high mark in Software Project Management.

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